CHEMISTRY ONLY

9 METALLIC BONDING

In this chapter we will discuss the bonding in metals and explain some of the physical properties of metals in terms of structure and bonding.



▲ Figure 9.1 Metals have properties that make them useful as construction materials.



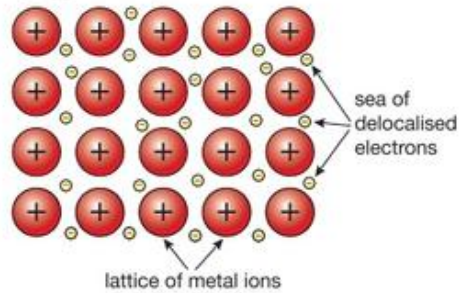
▲ Figure 9.2 Most metals are strong and all metals conduct electricity.

LEARNING OBJECTIVES

- Know how to represent a metallic lattice by a 2D diagram
- Understand metallic bonding in terms of electrostatic attractions
- Explain typical physical properties of metals, including electrical conductivity and malleability

METALLIC BONDING

Sodium is a metal. When sodium atoms bond together to form the solid metal, the outer electron on each sodium atom becomes free to move throughout the whole structure. The electrons are said to be *delocalised*. These electrons are no longer attached to particular atoms or pairs of atoms, instead you can think of them as flowing around through the whole metal. When a sodium atom loses its outer electron a sodium ion (Na^+) is left behind. A metallic structure consists of *a lattice (regular arrangement) of positive ions in a sea of delocalised electrons* (Figure 9.3).



▲ Figure 9.3 The metallic structure of a metal such as sodium

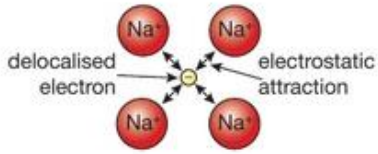
HINT

Different metals have different arrangements of ions in the lattice. Do not worry too much about this when drawing a diagram of a metal. As long as you draw the ions in a regular arrangement that will be fine.

HINT

When they come to write the symbol for a metal such as sodium in equations, students who know about metallic bonding sometimes worry whether they should write it as Na or Na^+. You write it as atoms, as Na. Thinking about the structure as a whole, the number of electrons exactly balances the number of positive charges. The metal as a whole carries no charge.

Metallic bonding is the electrostatic forces of attraction between each positive ion and the delocalised electrons. This holds the structure together.



▲ Figure 9.4 The electrostatic attraction between positively charged ions and negatively charged electrons holds the metallic structure together.

Metals have giant structures. There are no individual molecules and all the positive ions in the lattice attract all the delocalised electrons.

The ion formed by the metal depends on the number of electrons the original atom has in its outer shell. Thus all the elements in Group 1 form 1+ ions and all the elements in Group 2 form 2+ ions.

PHYSICAL PROPERTIES OF METALS

Most metals are hard and have high melting points. This suggests that the electrostatic forces of attraction between the positive ions and the delocalised electrons are strong.

In the case of sodium, only one electron per atom is delocalised, leaving ions with only one positive charge on them. The bonding in sodium is quite weak, as metals go, which is why sodium is fairly soft, with a low melting point for a metal. Magnesium has two outer electrons, both of which are delocalised into the 'sea', leaving behind ions that carry a charge of 2+. There is a much stronger electrostatic attraction between the 2+ ions and the delocalised electrons. This means that the bonding is stronger in magnesium and the melting point is higher.

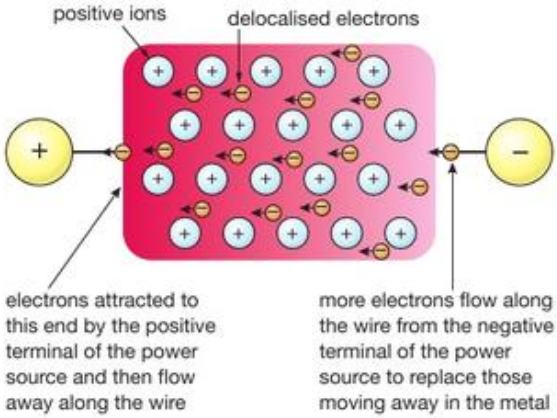
METALS CONDUCT ELECTRICITY

KEY POINT

In Chapter 7 we discussed ionic compounds and how they conduct electricity when molten or in aqueous solution because the **ions** are free to move. Ionic substances do not conduct electricity when solid because the ions are not free to move.
Metals conduct electricity when solid or molten because the delocalised **electrons** are free to move. It is important not to confuse this:

- ionic substances: ions move
- metals: electrons move

Metals conduct electricity because the *delocalised electrons are free to move throughout the structure*. Imagine what happens if a piece of metal is attached to an electrical power source.



▲ Figure 9.5 How metals conduct electricity.

METALS ARE MALLEABLE

Metals can be hammered into different shapes; this is what the word **malleable** means. When we apply a force to a piece of metal *the layers of positive ions slide over each other*. This does not affect the bonding in the structure; the positive ions are still attracted to the delocalised electrons.